Help your robot read colors

Name: ______________________________

Date: ____________________ **Period:** ____________________

Today I can...

Test paper colors and choose a setup that gives clear readings.

Words to know

Reading: The information a sensor gives you.

Unknown reading: The sensor cannot name the color.

Fair test: Change one thing at a time.

LOOK AT THE IDEA

Gather your materials

- Checked base with its downward-facing sensor
- Thin card about 30 cm by 12 cm
- Red, white and blue paper squares, each 8 cm wide
- Removable tape, ruler and pencil

Build a color-sensor test board

Build with the power off. Tick each step when it is done.

1. Tape the three paper squares side by side on the card: red, white and blue. Leave a small gap between them. Keep the paper flat, without shiny tape over the reading area.
2. Keep the robot’s wheels raised on firm supports. Slide the board under the sensor. Keep every motor stopped for this lesson.
3. Mark the board position that puts the center of red under the sensor. Do the same for white and blue. These marks help you line up each test.
4. Measure the gap from the sensor face to the board. Ask the teacher to check it against LEGO’s guidance. Record the gap; do not assume one height works for every build.

Sketch the setup. Label the parts you will check.

Finish the setup

5. Use the program below to show the color name. Test red three times, white three times and blue three times. Move the board away and back to its mark between readings.

6. Write down all nine readings, even wrong or unknown ones. Unknown means the sensor cannot name the color.

7. Change only one thing, such as room lighting or the sensor gap. Repeat all nine readings. Keep the paper and its position the same.

8. Choose the setup that gives the most reliable red and white readings. Record the gap and lighting. Keep the same paper for the stopping lesson.

- [ ] Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Start with all movement stopped.

2. Repeat 10 times: read the color, show its name, and wait 0.2 seconds.

3. Stop all movement and end.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.